

Totally Integrated Automation Portal

PLC\_1 [CPU 1214C DC/DC/DC]

PLC\_1

Project information

|      |       |        |          |         |  |
|------|-------|--------|----------|---------|--|
| Name | PLC_1 | Author | JohnWork | Comment |  |
| Slot | 1     | Rack   | 0        |         |  |

Catalog information

|                   |                    |             |                                                                                                                                                                                                                                                                                                                                                                    |                |                     |
|-------------------|--------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|
| Short designation | CPU 1214C DC/DC/DC | Description | Work memory 75 KB; 24VDC power supply with DI14 x 24VDC SINK/ SOURCE, DQ10 x 24VDC and AI2 on board; 6 high-speed counters and 4 pulse outputs on board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET interface for programming, HMI and PLC-to-PLC communication | Article number | 6ES7 214-1AG40-0XB0 |
| Firmware version  | V4.0               |             | False                                                                                                                                                                                                                                                                                                                                                              |                |                     |

Connection resources

|                   |   |                                            |    |                         |   |
|-------------------|---|--------------------------------------------|----|-------------------------|---|
| PG communication: | 1 | OP communication:                          | 1  | S7 basic communication: | 0 |
| S7 communication: | 0 | Maximum number of S7 connection resources: | 38 |                         |   |

Overview of addresses\Overview of addresses\Overview of addresses

|        |      |         |      |              |       |
|--------|------|---------|------|--------------|-------|
| Inputs | True | Outputs | True | Address gaps | False |
| Slot   | True |         |      |              |       |

| Type | Addr. from | Addr. to | Module          | PIP              | Device name                | Device number | Size    | Master / IO system | Rack | Slot |
|------|------------|----------|-----------------|------------------|----------------------------|---------------|---------|--------------------|------|------|
| I    | 0          | 1        | DI14/DQ10_1     | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 1  |
| O    | 0          | 1        | DI14/DQ10_1     | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 1  |
| I    | 64         | 67       | AI2_1           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 2  |
| I    | 1000       | 1003     | HSC_1           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 16 |
| I    | 1004       | 1007     | HSC_2           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 17 |
| I    | 1008       | 1011     | HSC_3           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 18 |
| I    | 1012       | 1015     | HSC_4           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 19 |
| I    | 1016       | 1019     | HSC_5           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 20 |
| I    | 1020       | 1023     | HSC_6           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 4 Bytes | -                  | 0    | 1 21 |
| O    | 80         | 81       | AQ1 x 12 bits_1 | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 3  |
| O    | 1000       | 1001     | Pulse_1         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 32 |
| O    | 1002       | 1003     | Pulse_2         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 33 |
| O    | 1004       | 1005     | Pulse_3         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 34 |
| O    | 1006       | 1007     | Pulse_4         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC] | -             | 2 Bytes | -                  | 0    | 1 35 |

Totally Integrated Automation Portal

### PLC\_1 [CPU 1214C DC/DC/DC] / Program blocks

#### Main [OB1]

Main Properties

General

|           |           |        |   |      |    |          |     |
|-----------|-----------|--------|---|------|----|----------|-----|
| Name      | Main      | Number | 1 | Type | OB | Language | FBD |
| Numbering | Automatic |        |   |      |    |          |     |

Information

|         |                              |                 |  |         |  |        |  |
|---------|------------------------------|-----------------|--|---------|--|--------|--|
| Title   | "Main Program Sweep (Cycle)" | Author          |  | Comment |  | Family |  |
| Version | 0.1                          | User-defined ID |  |         |  |        |  |

| Name     | Data type | Default value |
|----------|-----------|---------------|
| Temp     |           |               |
| Constant |           |               |

#### Network 1: Motor Control

```
graph LR; I0_0["%I0.0  
START"] --> Cmp[">=1"]; Q0_2_in["%Q0.2  
MOTOR"] --> Cmp; Cmp --> And["&"]; I0_1["%I0.1  
STOP"] --> And; And --> Out["="]; Out --> Q0_2_out["%Q0.2  
MOTOR"];
```

#### Network 2: Pilot Light 1 OFF (Red)

```
graph LR; Q0_0["%Q0.0  
PL1_OFF"] --> Out["="]; Q0_2["%Q0.2  
MOTOR"] -- R --> Out;
```

#### Network 3: Pilot Light 1 ON (Green)

```
graph LR; Q0_1["%Q0.1  
PL2_ON"] --> Out["="]; Q0_2["%Q0.2  
MOTOR"] -- R --> Out;
```

|                                                                                                      |  |  |
|------------------------------------------------------------------------------------------------------|--|--|
| Totally Integrated Automation Portal                                                                 |  |  |
| <div>PLC_1 [CPU 1214C DC/DC/DC]</div> <div>Technology objects</div> <div>This folder is empty.</div> |  |  |
|                                                                                                      |  |  |

PLC\_1 [CPU 1214C DC/DC/DC] / PLC tags / Default tag table [32]

PLC tags

| PLC tags                                                                          |         |           |         |        |
|-----------------------------------------------------------------------------------|---------|-----------|---------|--------|
|                                                                                   | Name    | Data type | Address | Retain |
|  | START   | Bool      | %I0.0   | False  |
|  | STOP    | Bool      | %I0.1   | False  |
|  | PL1_OFF | Bool      | %Q0.0   | False  |
|  | PL2_ON  | Bool      | %Q0.1   | False  |
|  | MOTOR   | Bool      | %Q0.2   | False  |



|                                                                                                  |  |  |
|--------------------------------------------------------------------------------------------------|--|--|
| Totally Integrated Automation Portal                                                             |  |  |
| <div>PLC_1 [CPU 1214C DC/DC/DC]</div> <div>PLC data types</div> <div>This folder is empty.</div> |  |  |
|                                                                                                  |  |  |



| Totally Integrated Automation Portal                                                                                                                                                                                 |         |                |             |         |                |             |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|-------------|---------|----------------|-------------|--|--|--|--|
| <div>PLC_1 [CPU 1214C DC/DC/DC] / Watch and force tables</div> <div>Force table</div> <table><tr><th>Name</th><th>Address</th><th>Display format</th><th>Force value</th></tr><tr><td colspan="4"></td></tr></table> |         |                | Name        | Address | Display format | Force value |  |  |  |  |
| Name                                                                                                                                                                                                                 | Address | Display format | Force value |         |                |             |  |  |  |  |
|                                                                                                                                                                                                                      |         |                |             |         |                |             |  |  |  |  |
|                                                                                                                                                                                                                      |         |                |             |         |                |             |  |  |  |  |

| Totally Integrated Automation Portal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |                |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|--------------|---------|----------------|--------------|---------|-------|------|--|-----------|-------|------|--|----------|-------|------|--|---------|-------|------|--|--------|-------|------|--|
| <div>PLC_1 [CPU 1214C DC/DC/DC] / Watch and force tables</div> <div>Tom_Watch_Table</div> <table><thead><tr><th>Name</th><th>Address</th><th>Display format</th><th>Modify value</th></tr></thead><tbody><tr><td>"MOTOR"</td><td>%Q0.2</td><td>Bool</td><td></td></tr><tr><td>"PL1_OFF"</td><td>%Q0.0</td><td>Bool</td><td></td></tr><tr><td>"PL2_ON"</td><td>%Q0.1</td><td>Bool</td><td></td></tr><tr><td>"START"</td><td>%I0.0</td><td>Bool</td><td></td></tr><tr><td>"STOP"</td><td>%I0.1</td><td>Bool</td><td></td></tr></tbody></table> |         |                | Name         | Address | Display format | Modify value | "MOTOR" | %Q0.2 | Bool |  | "PL1_OFF" | %Q0.0 | Bool |  | "PL2_ON" | %Q0.1 | Bool |  | "START" | %I0.0 | Bool |  | "STOP" | %I0.1 | Bool |  |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Address | Display format | Modify value |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
| "MOTOR"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | %Q0.2   | Bool           |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
| "PL1_OFF"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %Q0.0   | Bool           |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
| "PL2_ON"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | %Q0.1   | Bool           |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
| "START"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | %I0.0   | Bool           |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
| "STOP"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | %I0.1   | Bool           |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                |              |         |                |              |         |       |      |  |           |       |      |  |          |       |      |  |         |       |      |  |        |       |      |  |



| Totally Integrated Automation Portal                                                          |  |  |      |
|-----------------------------------------------------------------------------------------------|--|--|------|
| <div>PLC_1 [CPU 1214C DC/DC/DC]</div> <div>Traces</div> <table><tr><th>Name</th></tr></table> |  |  | Name |
| Name                                                                                          |  |  |      |
|                                                                                               |  |  |      |

|                                                                                                         |  |  |
|---------------------------------------------------------------------------------------------------------|--|--|
| Totally Integrated Automation Portal                                                                    |  |  |
| <div>PLC_1 [CPU 1214C DC/DC/DC] / Traces</div> <div>Measurements</div> <div>This folder is empty.</div> |  |  |
|                                                                                                         |  |  |

|                                                                                                            |  |  |
|------------------------------------------------------------------------------------------------------------|--|--|
| Totally Integrated<br>Automation Portal                                                                    |  |  |
| <div>PLC_1 [CPU 1214C DC/DC/DC] / Traces</div> <div>Combined measurements</div> <div><div>Name</div></div> |  |  |
|                                                                                                            |  |  |

|                                                                                                        |  |  |
|--------------------------------------------------------------------------------------------------------|--|--|
| Totally Integrated Automation Portal                                                                   |  |  |
| <div>PLC_1 [CPU 1214C DC/DC/DC]</div> <div>PLC alarm text lists</div> <div>This folder is empty.</div> |  |  |
|                                                                                                        |  |  |

PLC\_1 [CPU 1214C DC/DC/DC] / Local modules

PLC\_1 [CPU 1214C DC/DC/DC]

|                                                                   |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
|-------------------------------------------------------------------|------------|--------------------|-----------------|------------------|--------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|-------------------------|--|
| PLC_1                                                             |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
| Project information                                               |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
| Name                                                              |            | PLC_1              |                 |                  | Author                                     |               | JohnWork                                                                                                                                                                                                                                                                                                                                                           |                    |      | Comment                 |  |
| Slot                                                              |            | 1                  |                 |                  | Rack                                       |               | 0                                                                                                                                                                                                                                                                                                                                                                  |                    |      |                         |  |
| Catalog information                                               |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
| Short designation                                                 |            | CPU 1214C DC/DC/DC |                 |                  | Description                                |               | Work memory 75 KB; 24VDC power supply with DI14 x 24VDC SINK/ SOURCE, DQ10 x 24VDC and AI2 on board; 6 high-speed counters and 4 pulse outputs on board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET interface for programming, HMI and PLC-to-PLC communication |                    |      | Article number          |  |
| Firmware version                                                  |            | V4.0               |                 |                  |                                            |               | False                                                                                                                                                                                                                                                                                                                                                              |                    |      | 6ES7 214-1AG40-0XB0     |  |
| Connection resources                                              |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
| PG communication:                                                 |            | 1                  |                 |                  | OP communication:                          |               | 1                                                                                                                                                                                                                                                                                                                                                                  |                    |      | S7 basic communication: |  |
| S7 communication:                                                 |            | 0                  |                 |                  | Maximum number of S7 connection resources: |               | 38                                                                                                                                                                                                                                                                                                                                                                 |                    |      |                         |  |
| Overview of addresses\Overview of addresses\Overview of addresses |            |                    |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      |                         |  |
| Inputs                                                            |            | True               |                 |                  | Outputs                                    |               | True                                                                                                                                                                                                                                                                                                                                                               |                    |      | Address gaps            |  |
| Slot                                                              |            | True               |                 |                  |                                            |               |                                                                                                                                                                                                                                                                                                                                                                    |                    |      | False                   |  |
| Type                                                              | Addr. from | Addr. to           | Module          | PIP              | Device name                                | Device number | Size                                                                                                                                                                                                                                                                                                                                                               | Master / IO system | Rack | Slot                    |  |
| I                                                                 | 0          | 1                  | DI14/DQ10_1     | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 1                     |  |
| O                                                                 | 0          | 1                  | DI14/DQ10_1     | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 1                     |  |
| I                                                                 | 64         | 67                 | AI2_1           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 2                     |  |
| I                                                                 | 1000       | 1003               | HSC_1           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 16                    |  |
| I                                                                 | 1004       | 1007               | HSC_2           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 17                    |  |
| I                                                                 | 1008       | 1011               | HSC_3           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 18                    |  |
| I                                                                 | 1012       | 1015               | HSC_4           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 19                    |  |
| I                                                                 | 1016       | 1019               | HSC_5           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 20                    |  |
| I                                                                 | 1020       | 1023               | HSC_6           | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 4 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 21                    |  |
| O                                                                 | 80         | 81                 | AQ1 x 12 bits_1 | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 3                     |  |
| O                                                                 | 1000       | 1001               | Pulse_1         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 32                    |  |
| O                                                                 | 1002       | 1003               | Pulse_2         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 33                    |  |
| O                                                                 | 1004       | 1005               | Pulse_3         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 34                    |  |
| O                                                                 | 1006       | 1007               | Pulse_4         | Automatic update | PLC_1 [CPU 1214C DC/DC/DC]                 | -             | 2 Bytes                                                                                                                                                                                                                                                                                                                                                            | -                  | 0    | 1 35                    |  |